

Nash Equilibrium and Steady State

- **Nash equilibrium** (EK. Ch.6.4, Ch.6.5) is a profile of actions where each player chooses the best action against the actions of the other players (his neighbors).
- For any $q \in (0, 1)$ (for any $a, b > 0$ if interpret this as coordination games) and any connected network, all A and all B are Nash equilibrium (
- It is possible for A and B to coexist in equilibrium (example?).

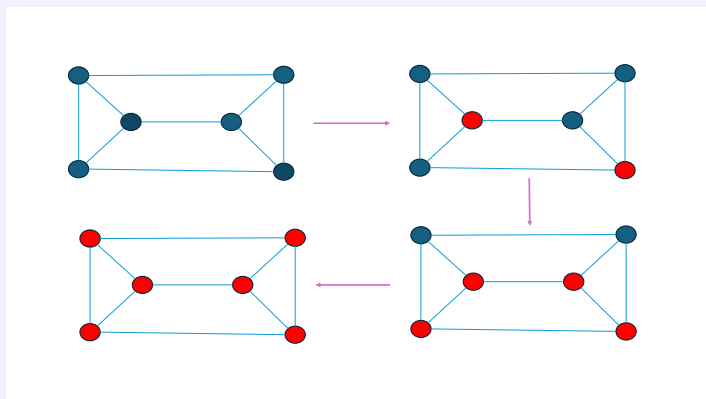
Cascading Behavior

Consider the following dynamical process:

- Everyone adopts B initially.
- A set of “initial adopters” adopt new action A for some reason (innovators, tech communities, payoff shocks...). They continue to play A every period.
- Every other player choose A if and only if the fraction of A players around them exceeds the threshold q (they play a myopic best response against their neighbors' actions in the previous period).

Initial adopters may cause a chain-reaction of A actions. When such a cascade occurs, the number of A s is increasing over time until the cascade stops.

Example: $q = 0.55$ and two nodes of initial adapters (red) $\rightarrow$ everyone choosing red in three steps (note that $b > a$. So, this is an inefficient change).



Does Efficient Technology always Spread?

- Suppose that A is a new superior technology ($a > b$) and hence $q < 0.5$. Does A spread to the whole society? How many initial adopters are needed for that?
 - ▶ Line?
 - ▶ 2 dimensional grid? (4-regular graph)
 - ▶ Star?
 - ▶ Clique?

What if two different players were initial adopters?

Cascades and Clusters

- When does a cascade of behavior stop? When does it replace B with A in the entire network?
- Intuitively, a tightly-knit group is resistant to a new behavior.
- A group of nodes is a **cluster of density p** if each node in the group has at least a fraction p of its neighbors within the group.
- Whether a cascade sweeps off the entire community or not crucially depends on the maximum density of all clusters.

Cascades and Clusters

- No member of a p -density group would adopt a new action A (even if everyone outside of the group adopts A) if $p > 1 - q$ (say, $q = 2/5$ and $p = 2/3$).
- On the other hand, if the density of every cluster is less than or equal to $1 - q$, then any small cascade of A would wipe off B from the society.
 - ▶ Suppose that, after a cascade started and stopped, there are some remaining players who are not converted.
 - ▶ Then the group of all such remaining players must be a cluster of density $p > 1 - q$. That is a contradiction.
 - ▶ Hence everyone should have adopted A when the cascade stops.

Application: Viral Marketing

- The choice of initial adopters affect how many nodes adopt new actions in the end.
- Consider marketers with limited advertising budget. Who should they target as initial adopters?
 - ▶ Those with high degree.
 - ▶ Those whose neighbors are more easily convertible (stronger incentive for adoption (as discussed later), low degree).
- If we know the distribution of consumer demographics (age, gender, race, locations) and statistics about across-within group connections & group-specific thresholds, then we can simulate such networks to find the right target.

Role of Weak Ties

- Local bridge conveys an information from one tightly-knit group to another. So, it may play an important role for information-based diffusion of behavior.
- On the other hand, a coordination-based diffusion of behavior tends to stop at local bridges.
- Information is more important for a diffusion of a low-threshold action. Coordination is more important for a diffusion of action with a higher threshold.

Heterogeneous Threshold

- We have assumed homogeneous people except for their positions in the network (and except for initial adopters).
- In reality, some people may be more/less willing to adopt a new behavior than others.
- We can incorporate such heterogeneous incentive by setting up individual payoff $a_v, b_v > 0$ with personal threshold $q_v = \frac{b_v}{a_v + b_v}$.
- More “influenceable” if a_v is larger and b_v is smaller. Such a node can be a good target.

- The result about cluster can be extended to this case.
 - ▶ A cluster V' is a **blocking cluster** if each node $v \in V'$ has strictly more than $1 - q_v$ fraction of neighbors within the cluster.
 - ▶ A started cascade of A wipes out B if and only if the remaining B players do not contain any blocking cluster.
- It is also possible to introduce weights to the edges and use the total weighted payoffs to determine which action to take.

Model with “Bilingual” Option

We consider another extension where each player has access to the third action AB : pay some cost to be able to coordinate with both A neighbors and B neighbors separately.

- Think of A and B are incompatible technologies and AB as a “bilingual” technology that is compatible with A and B .
- Payoff a for (A, A) and b for (B, B) (and 0 for incompatible choices) as before. AB costs $c > 0$ but works as A against A and as B against B for each coordination game as follows.
 - ▶ If (A, AB) or (AB, A) , then both players get a .
 - ▶ If (B, AB) or (AB, B) , then both players get b .
 - ▶ If (AB, AB) , they coordinate on a superior technology. Assume $a > b$.

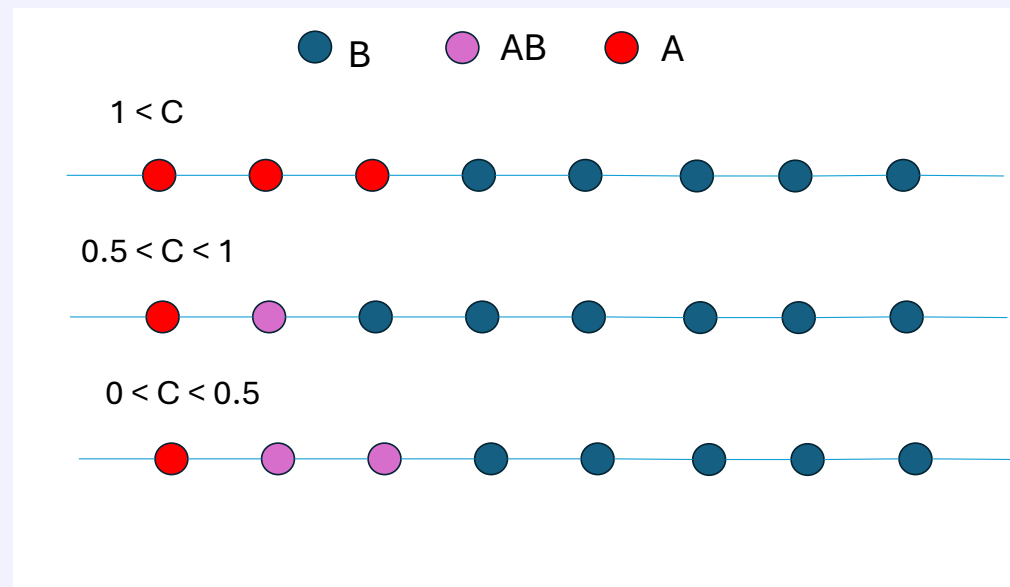
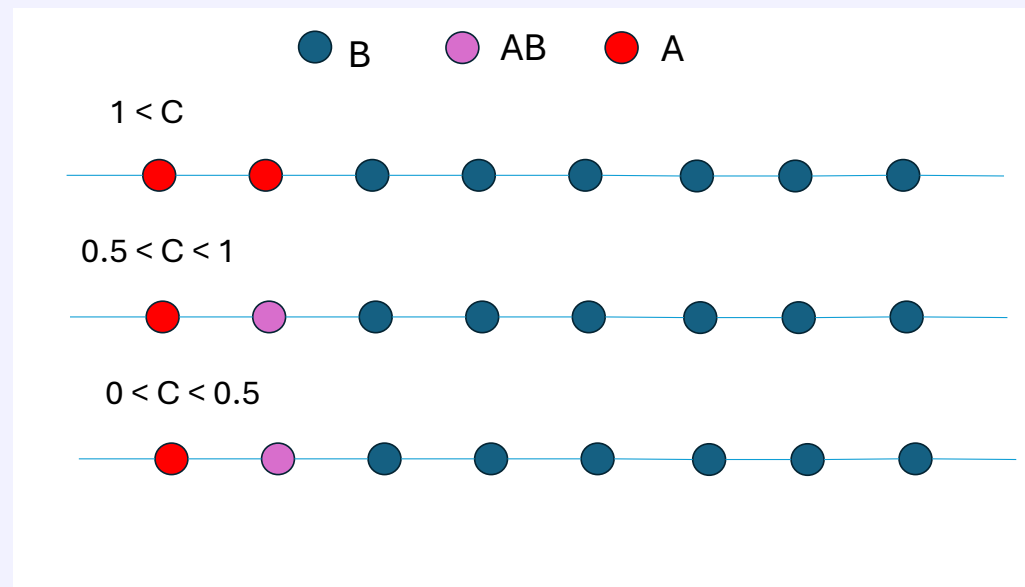
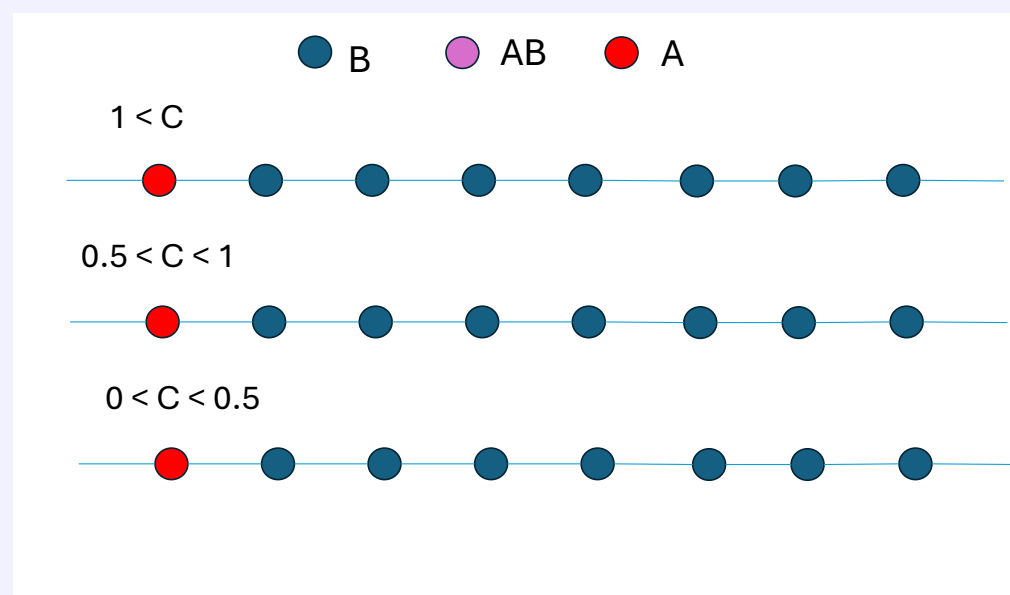
There are two effects of AB .

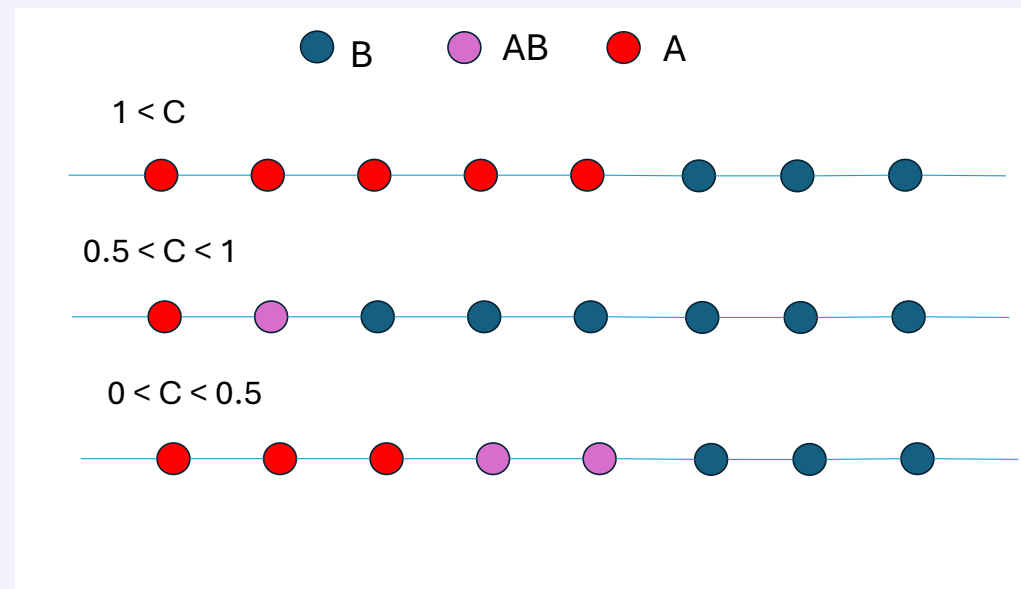
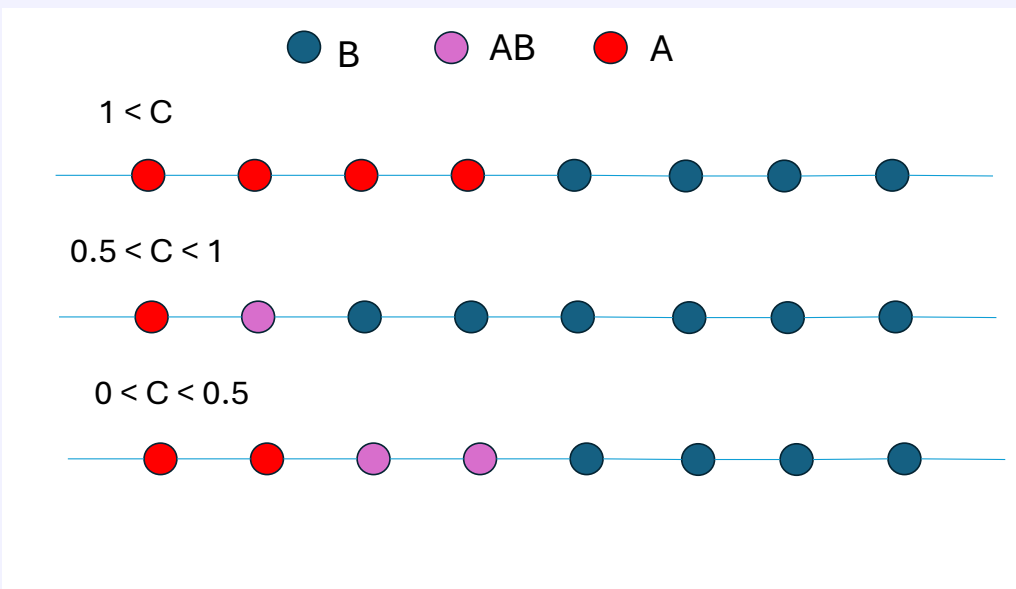
- Previously a player needs to give up B -neighbors when playing A and vice versa. Now they can coordinate with both A -neighbors and B -neighbors (with cost). This makes an introduction of A to the society easier.
- On the other hand, B players have a stronger incentive to continue playing B against AB than against A , because they know that AB is willing to coordinate to play B against her.

As an example, assume $a = 1.5$, $b = 1$, and a line network. Start with one initial adopter. The incentive for a node between A and B or between AB and B is important.

There are three cases depending on the cost size.

- For the high cost case ($1 < c$), the best response for a node between A and B is A .
- For the moderate cost case ($0.5 < c < 1$), the best response for a node between A and B is AB , but the best response for a node between AB and B is B .
- For the low cost case ($c < 0.5$), the best response for a node between A and B or between AB and B is AB (and the best response between A and AB is A).





Summary

- A basic model of diffusion of behavior based on the coordination incentive.
- A cascade of a new action stops at a Nash equilibrium. It may or may not sweep away the old action completely depending on the threshold and the network structure.
- We considered various extensions of the basic model.

Suggested reading. EK Ch.19.1 - Ch.19.5, Ch.6.1 - Ch.6.5